

Sondaki Amfinin lineerliği en büyük olmalı

$IIP_3 > IIP_2 > IIP_1$

$IIP1_3 > IIP1_2 > IIP1_3$

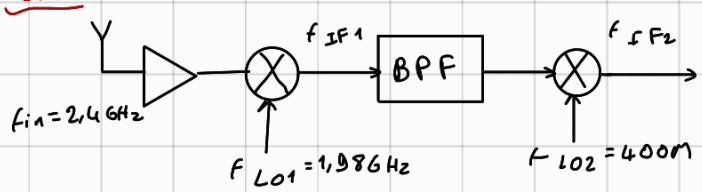
$$x_{Lo}(t) = A_{LO1}\cos(2\pi f_{o1}t) + A_2\cos(2\pi 2f_{o1}t) + \dots + \cdots\cos(2\pi n f_{o1}t)$$

mixing spurs

$f_{in} \pm m f_{LO1}$ (expected spurs freqs)

$f_{in} \pm m f_{LO1} \pm n f_{LO2}$ → expected mixing spurs in dual conversion

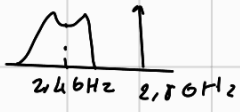
Ex:



$f_{IF_1} = |f_{in} - f_{LO1}| = 2,46 - 1,986 = 420M$

$f_{IF2} = |f_{LO1} - f_{LO2}| = 420 - 400 = 20M$

f_{LO2} (2nd harmonic) = 800 MHz



f_{in} (2,8GHz) − f_{LO1} (1,986Hz) − f_{LO2}(2nd) (800MHz) = 20MHz

n / fRF	dBc (carriere göre) 0	1	2	3	4	5
0						
1		0	-20	-30	-40	
2						
3						
4			-60	-80	-70	

$f_{RF} = 2$ GHz

$f_{LO} = 1,9$ GHz

$f_{FF} = 100$ MHz

Dual Conversion receivers

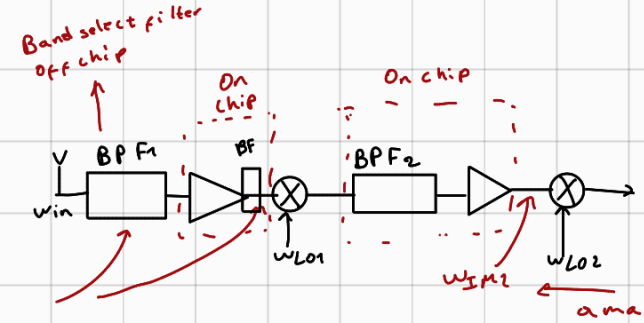
Disavantages:

- 2 tane local oscillators
- More mixing spurs
- Several passive filters

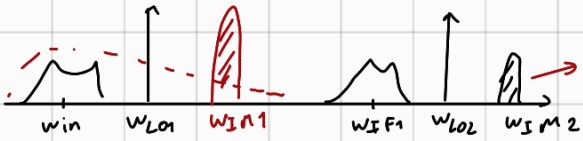
Advantages:

- Robust
  clear choice for high sensitive radio receives
- High dynamic Range
- Power consumption is lower
- Secondary Image frequency





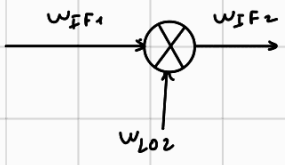
ama buradan havadan girişten gelemez



havadan sızıntıyı engellemek kolaydır
sistem tasarımındaki kat veya sistemden ortaya çıkabilir.

Removing Secondary Image

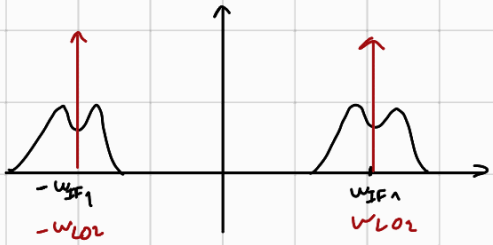




Zero-IF (Homodyne)

$$w_{IF2} = 0 = |w_{IF1} - w_{LO2}|$$

$$w_{LO2} = w_{IF1}$$





eğer simetrik değilse



non-symmetric ⟶

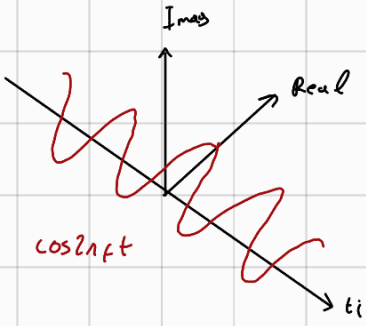


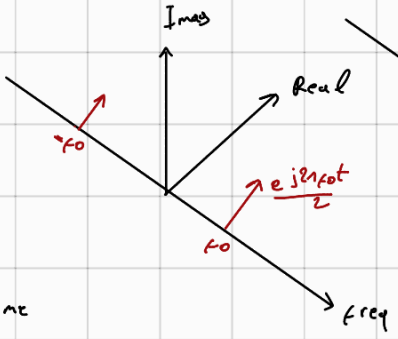
Quadrature Signals & Quadratic Down conversion

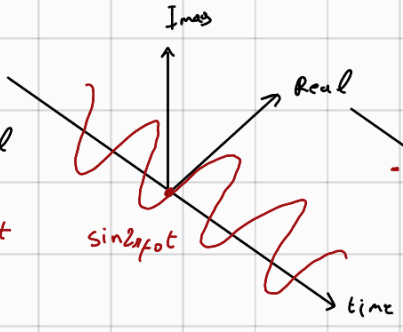
$$\cos 2\pi f_0 t = \frac{e^{j2\pi f_0 t}}{2} + \frac{e^{-j2\pi f_0 t}}{2} \xrightarrow{\mathcal{F}} \frac{1}{2}\left[\delta(f-f_0)+\delta(f+f_0)\right] \longrightarrow \text{real}$$

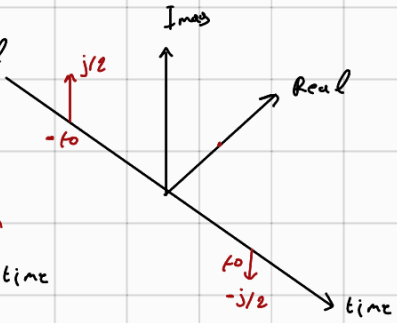
90° phase shift ↓

$$\sin 2\pi f_0 t = j\frac{e^{-j2\pi f_0 t}}{2} - j\frac{e^{j2\pi f_0 t}}{2} \xrightarrow{\mathcal{F}} \frac{j}{2}\left[-\delta(f-f_0)+\delta(f+f_0)\right] \rightarrow \text{Imag.}$$









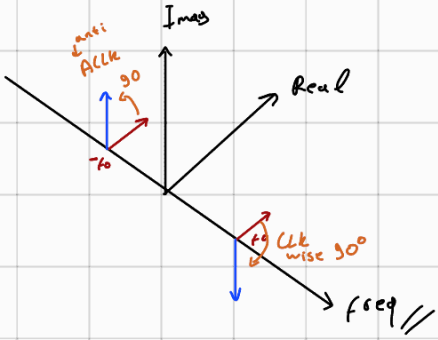
$$\cos(\omega_c t - 90°) = \frac{e^{j(\omega_c t - 90)}}{2} + \frac{e^{-j(\omega_c t - 90)}}{2}$$

$$= \frac{e^{j\omega_c t}}{2} e^{-j90} + \frac{e^{-j\omega_c t}}{2} \cdot e^{j90}$$

$e^{-j90} = \cos 90 - j\sin 90 = -j$

$e^{j90} = \cos 90 + j\sin 90 = j$

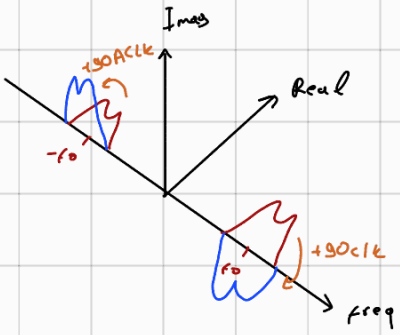
$$= -j\frac{e^{j\omega_c t}}{2} + j\frac{e^{-j\omega_c t}}{2} = \sin(\omega_c t)$$

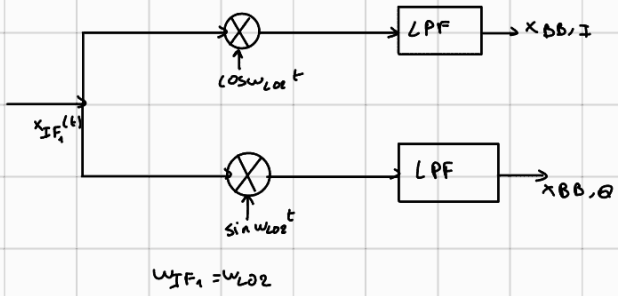


Positive frequency +90° Clkwise

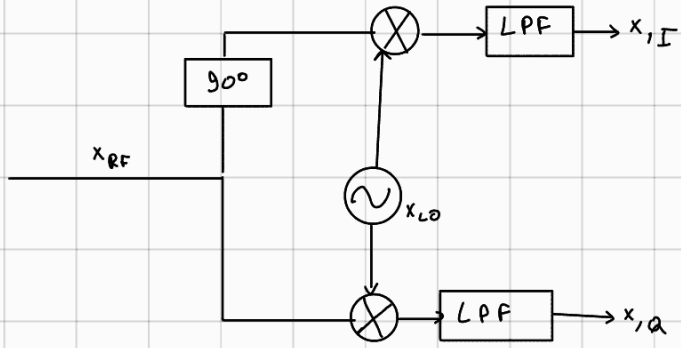
Negative " +90° AClkwise

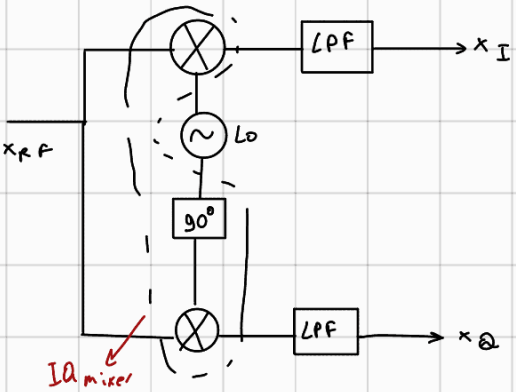
Demodulator



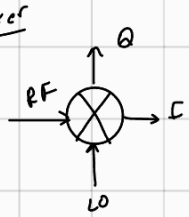


$\omega_{IF_1} = \omega_{LO2}$

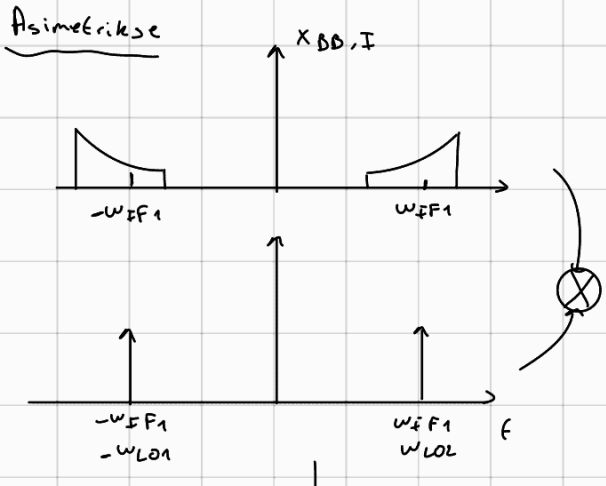


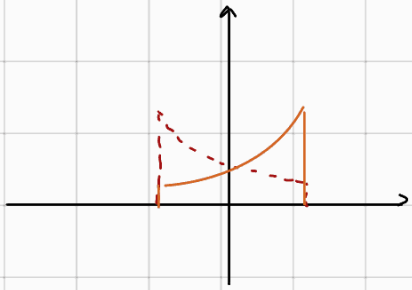


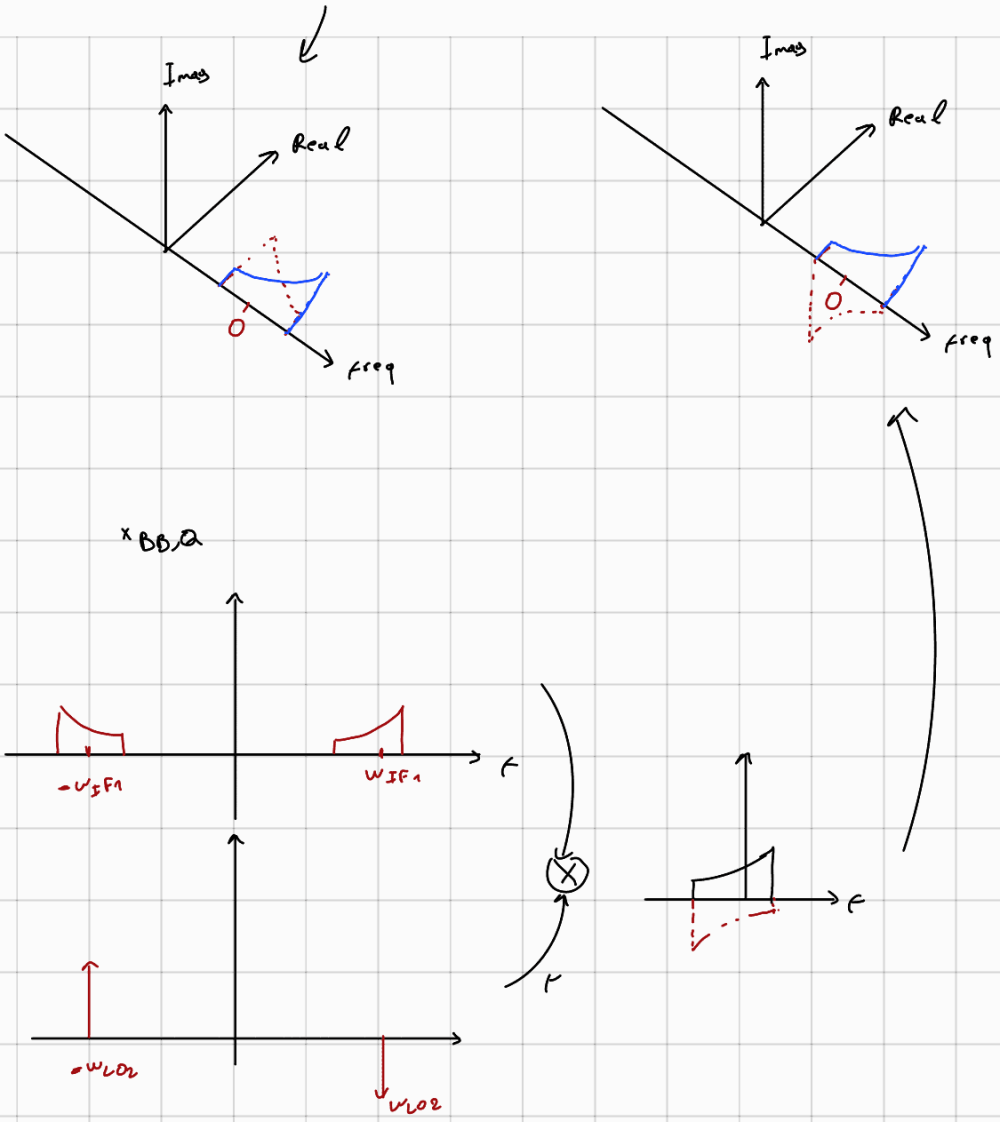
IQ Mixer

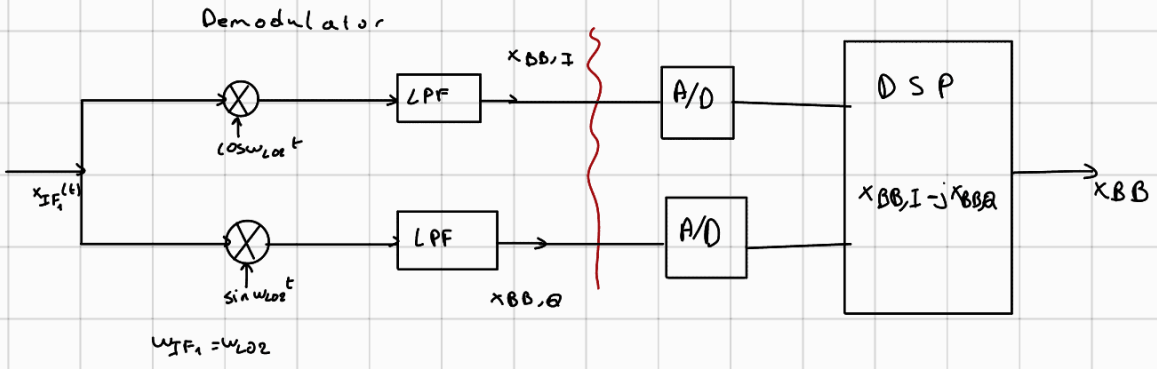


Asimetrikse

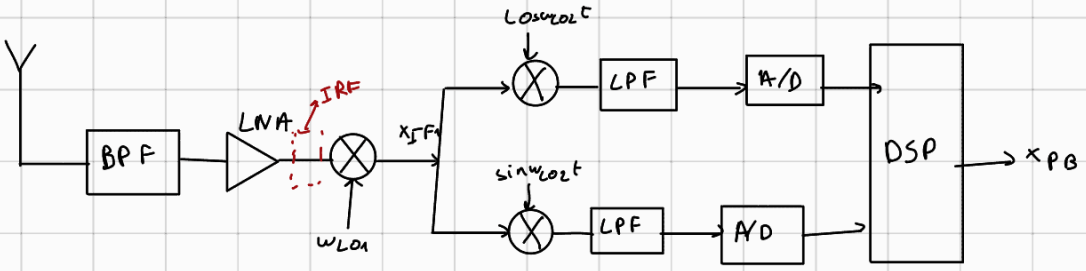


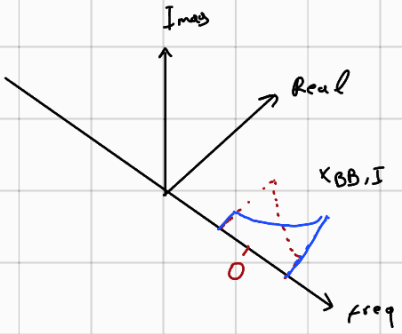




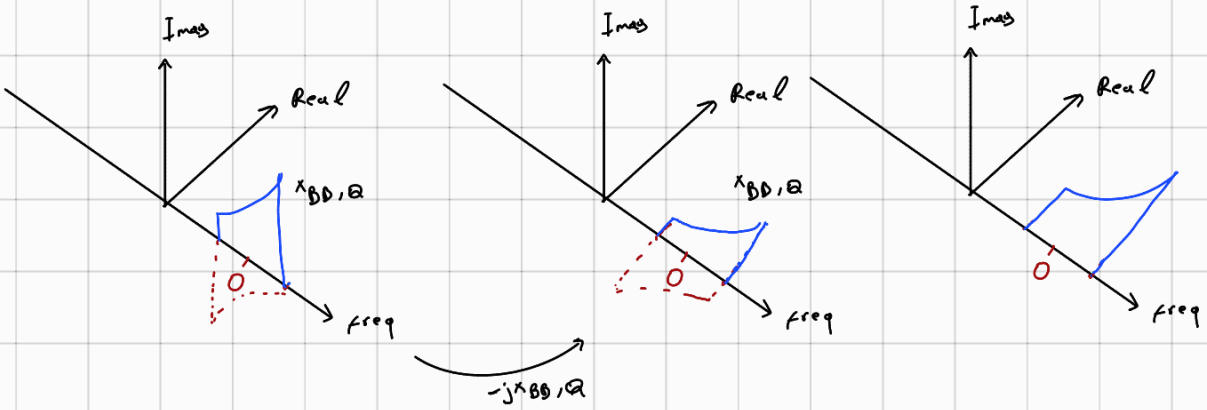


Zero - (Second) IF Heterodyne Receiver

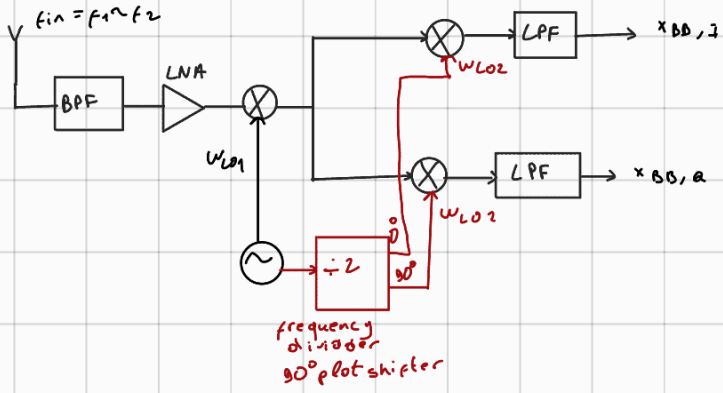




$$x_{BB,I}(t) - j\, x_{BB,Q}$$



Sliding IF Receiver

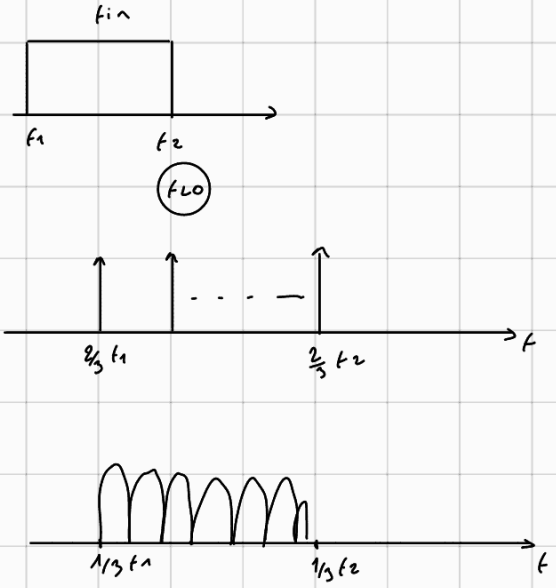


$f_{LO2} = \frac{1}{2} f_{LO1}$

(Zero-IF) $|f_{in} - f_{LO1} - f_{LO2}| = 0$

$f_{in} - f_{LO1} - \frac{1}{2} f_{LO1} = 0 \quad \Rightarrow f_{LO1} = \frac{2}{3} f_{in}$

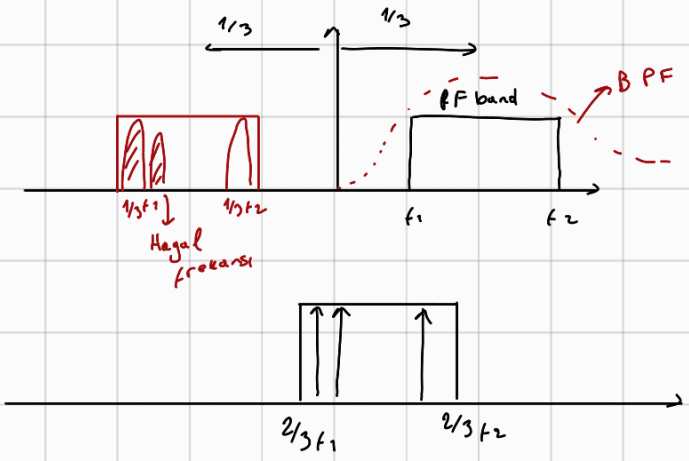
$\Rightarrow f_{IF1} = \frac{1}{3} f_{in}$



x_{IF2} = Zero IF

BB

DC

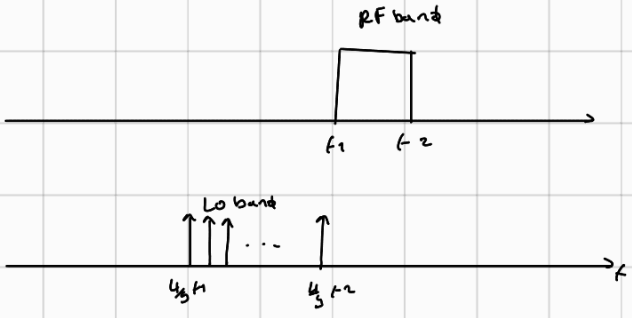


Both BPF and LNA reject the Image

Ex: $f_{LO2} = \frac{1}{4} f_{LO1}$

$|f_{in} - f_{LO1} - f_{LO2}| = 0$

$f_{in} - f_{LO1} - \frac{1}{4} f_{LO1} = 0 \rightarrow f_{LO1} = \frac{4}{5} f_{in} \rightarrow f_{IF1} = \frac{1}{5} f_{in}$

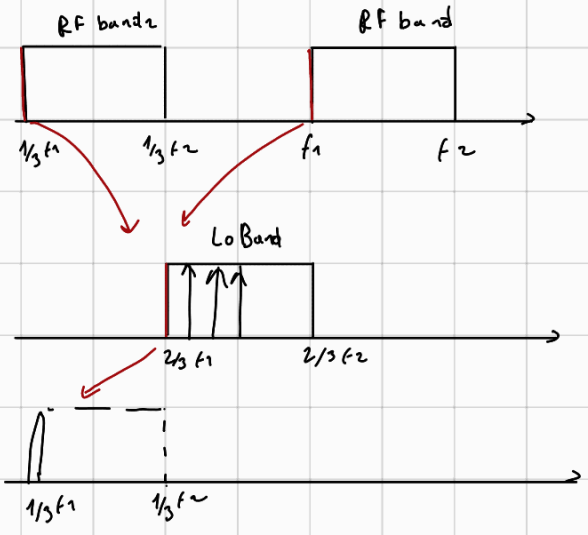


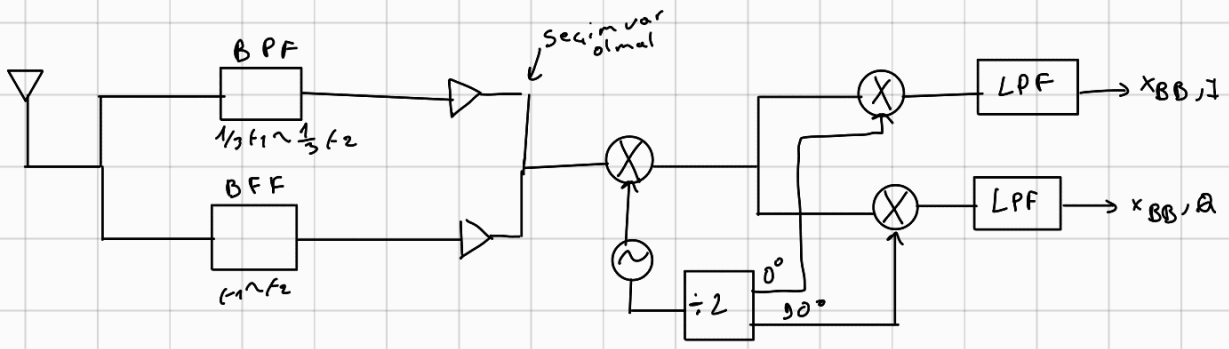
RF bandına LO gaklaşınca image rejection zor olur

/8 gibi bir şey kullanma image rejection zor olur



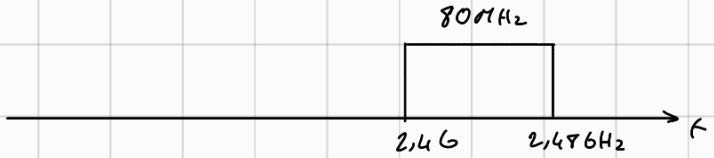
Dual Band Receiver



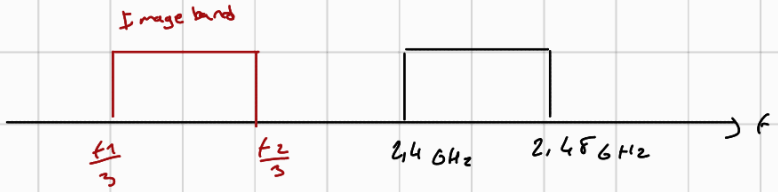


↓ Wifi standart

802.11g ÷2 = ?
÷4 = ?

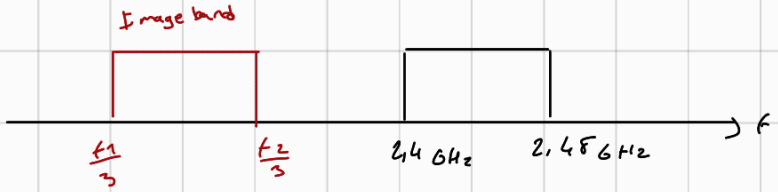


÷2



0,80 0,827G
800MHz 827MHz

GSM band → Somany application in this band. It's not quiet.



÷4

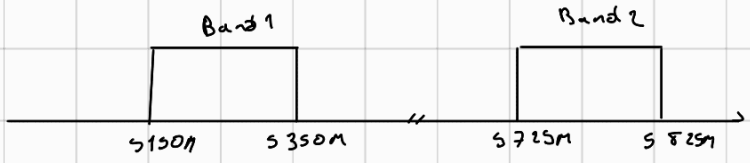
1,46 1,486Hz

The probability of image is low

Bu daha iyi

Galiema sorusu

Ex: 802.11a, ÷8 structure sliding Receiver



1) Image Band?

2) Mixing spurs in results of mixing with 3rd harmonic of the first LO & 3rd harmonic of second LO

$f_{in} - 3f_{LO1} - f_{LO2} = ?\ 0$

$f_{in} - f_{LO1} - 3f_{LO2} = 0$